

Single market trend has a very good year and that is right because you've seen these major trends in like equities and gold, silver and markets that we've mentioned. It deserves to be large. It deserves to be moving the needle. It deserves a larger footprint in our portfolio because that's the outlier trade that's working. And other trades that haven't, or other markets that aren't trending, that aren't doing well, that therefore as a result aren't as large in our portfolio, why should we increase dumb? It's kind of like adding to losers. and taking away from winners, which is the exact opposite of a trend following strategies that is keeping losses small and letting winners run. I am joined today by Moritz Haydn and Moritz Siebert of Takahe Capital. Moritz, thank you both for being here today. Thank you for having us, Max. Thanks, Max. So we have been connected for a long time, going back to the real vision days. I've gotten to follow along with Takahe and your growth. You guys have been relentless, in trying to bring forth quantitative, diversifying strategies, really for absolute return purposes. Mostly trend following, but I know that what you do is a little bit more than trend. We'll get into those nuances. But I want to start there. Trend following has worked this year. It worked a lot better at the beginning of the year when gold was really trending. And it's one of the things that I have always found interesting. is sometimes you get a trend that is so big and so strong in a large enough market, it can kind of propel the entire asset class forward. And it really felt like gold was doing that at the beginning of the year. And so I'm interested in when trend following works, is it that everything is trending or that you really only need one or two big trends in a year to make the strategy perform the way it's supposed to? It's great when you have a lot of trends happening at the same time. We can follow trends and capture trends in a diverse set of markets, and they all trend, then that's great. Then we're going to have a fantastic time with trend following strategies. But more often than not, what you see is that you have some markets trending, like a smallest subset of your portfolio, and many markets inside your portfolio not really trending. Most of the trades that we initiate actually become small losing trades, which means, you know, we broke the market for position either long or short. It doesn't work. We take a small loss. It's appropriately sized and we move on. We like, you know, don't cry about that. It's just the nature of our business. But a couple of these trades,

they go on and go on and they become large and successful big trending trades. And these trades can become so.

large or so good in terms of profits that they will cover the losses of the many small losing trades that we had before and then make a some money on top of it. And you just never know which markets that's going to be. You know, earlier this year was big moves in gold and silver. Then with the onset of the Iran war, you had big moves in the petroleum markets that's, you know, Brent and WTI and heating oil and gas oil. You have trends in the agricultural markets, you know, just absent flows. You never know what it's going to be. Well, the gold and silver trend feels like more of a classic building trend where it's going for months and months and months. And not that petroleum hasn't trended, but it's definitely been rocky with a lot of job owning from the president. You're seeing a lot of gapping in the market, both up and down, whether there's events that are that are causing the market to gap up. And so I'm interested in how trend following strategies differ when you have sort of a smooth trend like gold, not that we didn't have our gap days in gold and silver, but it just feels like oil has been much more volatile, at least commodities tied to oil. So I'm interested in how your, your models treat those two different types of trends. So if you have a lot of volatility without direction, without the underlying trend being honored, then you have a big risk of being kicked out of your position. We call that a whipsaw, right? So you get into a trend, say you're buying oil, and then because of whatever announcement, it could be a, it's no longer a tweet, is it? It's a truth. So how do you call it when there's something on truth social? Maybe it's a truth. I don't know. The truth. It's the truth. That's all it is. It must be the truth, right? Yeah. And then, you know, oil moves, whatever, like up 15 or in our case, like worse down 15 bucks, right? And that could kick you out of the position. So you're losing your long position because of this very quick move. And then maybe just, you know, gold, oil goes back to where it was before. So that can happen now. But there is, there is oil isn't oil. I mean, in the sense that there is a futures curve and you can be positioned at the front of the futures curve or at any other. point of the futures curve that's liquid and supports your liquidity needs. In our case, for example, we didn't have front month exposure, which I presume was the April contract in WTI at least when the Iran war started. And that contract,

like that very short dated part of the curve, had the biggest moves. It has most of the beta to flat price, if you will.

And that is where he saw most of the action, like the front part of the curve went into almost record backwardation in a very swift period of time. But if you have exposure more to the dated part of the futures curve, say December of 2026, which is what we had, there's less volatility at that part of the futures curve. There's less of a reactive movement to any tweet or truth posting because, you know, The thing could be over by December or by November when that contract goes into expiration. Maybe prices will still be higher then, but maybe the conflict itself is over. So there's less of that access volatility being realized at that more dated part of the curve. And that is quite natural or standard for any of these markets. The further down the curve you go, the less volatility, the less beta you have to the front price action. It has consequences pros and cons. The pro might be that you're not losing your position in the example that I've just explained, right? Because you don't have that whipsawing feature. The con is that you don't get as much P&L. You know, if the market moves up and up and up and up and up and you're along the April contract in March or in the end of February, then you know that is where the money is made from the long side. The December contract will make you some money by holding a long position and it did make us some money from that long position, but absolutely not as much as a April contract position on the long side would have made us. Is it rare for you to get into the back end of the curve instead of the front end of the curve? Or is that by design? Are you specifically trying to find lower volatility trades in something where you know that it's highly likely? Is there any discretionary aspect to this? Or is it purely The model says the strength of the trend is at the back end of the curve or volatility adjusted. That trend is stronger. How much discretion goes into choosing what part of the curve? Yeah, there's definitely some discretion in the model building step, right? We have different models and different models might target different parts of the curve for different reasons. For example, the one in oil at Mordsmann sheet being out further on the curve. But it's it's not us kind of like getting a signal for example for oil and at that step we decide okay we're going in at the front or the back because volatility is currently high or low or whatever it's by definition already in the beginning we might have a model that is targeting the front of the curve

we might have a model that is targeting the back of the curve or even something like we do the spread trading where we actually trade some sort of breakout for example on different different contracts on the curve actually which

Again, we have totally differently because when we designed the models, we treated these different parts of the curve as their individual time series. So you might think that we have not only one oil idea or one oil time series in there, where the model does something with it, but we have maybe six different variations of oil, which look a little bit different, right? Because we are sitting on different points in the curve. And that adds to the diversification we can get overall in our systems because although they are kind of like bound together in some way they react differently. This is something as a machine really goes in at the front and usually isn't changed. So we do not we do not change the system in terms of like maybe now we are thinking of instead of in the front or back and go into the third contract or something. We don't kind of like have that discretion when the trade really comes about. So it's at the point of building models. And I think that that is an interesting topic to go a little deeper on because when you pull up, whether it's trend following ETFs, mutual funds, if you're looking at the indexes, there is quite a dispersion in returns in months over years and certainly over the long run. And it's always been fascinating to me because it. Yes, you can differ somewhat on the definition of what is a trend. But at the end of the day, like we kind of all know it when we see it. And so that has always just been fascinating to me. So can we talk a little bit more about the customization of trend following strategies and how very different they can be? And maybe what is sort of the classic trend following look like? And then what are the nuances that you guys have implemented to make yours different? Yeah, it can be very very different max. So the stuff that goes in at the front is the portfolio of markets that you trade You can run a trend following system on just the S&P 500 or on any other single market on a standalone basis, you know Yeah, it would be trend following it probably would be the best trend following system for sure It wouldn't be the best trend following system because you're lacking diversification, but In our case, we're trading a portfolio of 100 markets and that portfolio is very unique. I don't think that any other trend following manager trades the exact same portfolio as we do. So that is very important and it creates a lot

of differences between managers. You have managers that trade a very small number of markets, say 20, 21, 30 markets, 35 markets, something like that. You have managers that are very commodity heavy and they don't trade the financials. They stay away from

you know the equity indices or the bonds for whatever reason. And then you have trend following managers that trade 400 and 500 and 600 markets and they include cash equities and OTC trades and interest rate swaps and very exotic currencies and stuff like that. So you can already see like from what you put in at the front into the engine, if you will, into the models, into the systems that provide trading signals. If there is a fund that trades propane gas and Japanese power markets and New Zealand power markets and California carbon allowance, that's going to be very different than a, say, US ETF using a trend following system on just the major macro markets such as the S&P 500, the 10 year note and crude oil and markets such as that. So that creates a big difference. Then second up is the types of models and their trading speed. So what models are you trading? There's different types that could get you into a trend following position breakouts and moving averages and regression lines and time series momentum. There's different ways of exiting positions, trailing stops and other signaling techniques that would make you reverse a position or get out of a position. And then Depending on how you set these boundaries and these parameters, your system can be short term, medium term, long term. All of these are very subjective terms. There is no objective measurement for, oh, this is medium term. It's kind of like, yeah, well, you know, in our case, if you have, say, a holding period of an expectation, or if a trade is more like, you know, 100 days in terms of a look back window, we're looking back more than 100 days. And probably at that point becomes medium to a long term if you're looking back 30 days. or 20 days, just a month, that's probably a short term trading system. You will have much more action, much more in and out, much more whipsaw, much more frequent change of positions. And that will then also change or produce a different return profile. And then lastly, but also very importantly is what's the volatility that you're trading at? You have some funds trading at very high volatility or generally higher volatility and I would put us in that bucket. What is that? Can you quantify that with an annualized volatility? Yes, like 25% to 30% roughly. We don't target any number there. We're not

forcing our return stream to come out at 25% or 30% vol pretty much exactly. But it's on average going to be in that range. And you have funds that trade at even higher volatility than that and most funds. And that is really the...

the vast majority of CTA trend following funds and definitely the vast majority of assets traded in accordance with trend following models. There are more and kind of like the eight to 12% I would say volatility range because that is for institutional investors feel at home. That's kind of like, you know, easy to stomach, no big damage being done at that level of volatility. Right. But clearly a fund, even if you have the same markets, the same trading speeds, same, same, same, same. One trading at 12% volatility and the other at 30, you know, creates a big difference. And that is especially then also true if and when it's not if it's when these programs go into drawdowns and you know, drawdowns are a natural byproduct of trend following trading of any trading strategy, I presume. I mean, you most of the time you are in a drawdown and only very rarely do you make a new equity high, but you know, a fund trading at a higher level of volatility and expectation will have a higher drawdown. And when you are in these periods of a higher drawdown, you have the volatility drag, right? If you're down, we're not down 50%, but just for the ease of calculation and the simplicity of the math, if you're down 50%, you need to make 100. A CTA or trend following fund trading at a 10% volatility level is, it's not an impossibility for that fund to be down 50%. But it's just much more unlikely for that fund to be down 50% relative to a high volatility fund. So they will have less of a volatility drag. All of that creates differences. I like to say, and you know this, you're your podcasters yourself. Podcasts are not mediums for nuance. So my question is, what is the best trend following strategy and why? What is the best and why? More it's just clearly as a question for you. Probably isn't the best but like we can we can probably answer what's the most classic one I would say if you look at the the classic one they would always say like donkey and channels with Classic stop loss, right? This is kind of like a very classic trend following strategy and then the newer ones like you were once in terms of like Which we see a lot in another fund was maybe more like exponentially where it will be average across different speeds and then dynamic position sizing, aka target volatility, which is something they have always a lot of discussion between

the two types. I think one of is maybe the more old school group, the donkey and channels group, and then the target world group as a kind of newer kids on the block, because they come from a different background. And there has been a lot of discussion between the two groups as well as to what is kind of like the, yeah, the true, or maybe not the true, but maybe the best form of doing trend following right as an ongoing discussion. And there is no

not definitely the answer to that because it's so different in terms of like what's the return profile both ideas do create and what it is you get because one thing is like the very old-school trend following trades at Heiswald creates drawdowns yeah and it's choppy but it is also these outliers most like if you're cuckoo and put for your golden ever big position In that, and you allow that to grow over time, even though volatility kicks in, because you're not readjusting the position, this will have a big impact on your portfolio, and you'll get these alkali trades. On the other hand, there's dynamic wall guys who say, well, no matter, probably they might get in at the same time, into the same trade. So I'm always saying that the methodology for entry probably doesn't matter that much. There's some better ones, some really bad ones, maybe, but in between it's very similar, at least on our time frame. But then how do you positionize the thing over time? Maybe you're at entry or do you kind of like adjust it? That plays a big role. And for example, dynamic ball would touch a position and volatility rises, even if the movement is in your favor. So kind of like your long cocoa and volatility goes up, keep going through the rule, but you're targeting a fixed kind of like volatility contribution for your portfolio, you would dial down the position, take pro-giant. This is just two very different approaches and they create very different outcomes in terms of the return distribution. And I would say, for example, the dynamic world thing is very often easier to stomach for people because it creates a smoother profile. It has a higher sharp ratio, for example, while the old school methodology has its roughness, has a low sharp ratio usually that is heavily skewed, but nonetheless creates these enormous strengths. And we are more in the old school camp. Even though it's an oversimplified version now that I mentioned, but still the nature of things and how we trade it, how we see it is more in that direction. It's funny, you say that everybody can get in at around the same time, but it's how you size it. One of my favorite people to

read every year, Harley Bassman, the convexity maven. Every single one of the pieces that he finishes is with sizing is always more important than entry. Let's talk about sizing and why, I presume, based off of what you told me about most people's vol and your vol, that you are not doing dynamic sizing. You're letting the market determine the size. The market's volatility and our appetite for risk, like how much we would like to risk or how much our systems risk per trade, per unit, so to say. And you're right. And Harley Bassman is correct with his quote that

position sizing is more important than entry level, especially for trades that you're going to be holding longer term. Like that statement is not true if you're doing intraday trading, right? Or you're doing like, oh, I'm trading a five minute bar with the expectancy of being out in 15 minutes, you know, at these types of frequencies, obviously your entry and exit level are pretty much all that matters. But for us, with long-term trend following systems, if we do catch a trade and, you know, every once in a while we do, like long cocoa and short cocoa and short orange juice. And, you know, some of these positions, some of these trends, they become outliers and they last for a long time. We can hold a position for more than two years and it did happen and it does happen. And when you have that type of a timeframe, it doesn't really matter whether you entered a day later a day earlier or a week later or a week earlier or 10 days in the biggest scheme of things that's not going to be making or breaking that trade, but what will. Affected much more is like how many contracts did you buy or sell when you initiated the position and yes so sizing is very important sizing is.

Appropriate sizing like. You know, you don't want to have too little. You don't want to have too much. You want to have a diversified portfolio. And whatever happens, if the trade goes against you and it becomes a losing trade, which happens quite frequently, you will never want that trade to become a big loser because big losers are assets. They're really bad for your portfolio. You know, they have a larger footprint than that. start dominating the portfolio in the returns, you don't want that. You want the winners to produce the volatility and not the losers. What is a big loser? A big loser is something that should never happen. But on a quantify, quantify a big loser for me, is that 200 basis points of total capital? Is that 500 basis points? 200 basis points would be a massively big loser for us. We're trading a portfolio of 100 markets if we had, say, I mean. It's very unlikely for that to happen.

But if we initiated 100 positions and we risked 200 basis points on each of them, we could be losing 200%, which is more than the money that we have in the fund. So now we're risking a much smaller fraction of our close trade equity. Think of this as our bankroll or cash on hand, if you will. A much smaller fraction than 200 basis points on a trade.

For the simplicity of the calculation, let's just assume that we risk a hundred basis points, right? Then that is the expected loss on that trade. That is the money that we expect to lose if the trade hits its initial stop. In a second, I'd like to explain that we could on a giveback, in a giveback scenario where we sit on open trade profits and then the trade reverses against us, we can lose more than that. That is a different discussion to have because it happens at a different point in time of the trade usually. But yeah, so if we go into the initial stop, and my example, if we risk 100 basis points, then that is the loss that we would suffer. Now, is it going to be exactly 100 basis points that you lose? Very unlikely, but it should be in the proximity of that level. Why is it unlikely? Well, First of all, it depends on do you have a stop working for you in the market through your broker? Like, is there a resting order with your broker that would hit, would trigger a closure of your position in the market without doing anything? Then, yeah, you will then be very close, I presume, to your 100 basis point loss. net of slippage net of commissions net of bid offer there's always a little bit of an inconsistency there right it cannot be a hundred percent precise in our case we don't do that like our system will tell us oh you've hit your initial stop by the way it happened yesterday you want to get out of the position but we're getting out of the position with a delay you know say today so there's a could be a 24 hour delay or it could be a weekend in between. And then you know we accident position now between these two time points something will happen to the market. it could obviously stay flat and not move, and we get out with a hundred basis points loss, but it could be 120 basis points loss, it could be 110 basis points loss, it could be an 80 basis points loss. It works in both directions, but on average, if we go through all these trades, sometimes that works for us, sometimes that works against us, it's kind of like, yeah, the 100 basis points, if there is no discontinuity or gap event, which obviously could also happen. I mean, imagine you want to get out of oil on a Friday, and you don't execute it on a Friday, you do it on a Monday, and then some geopolitical event materializes over the weekend, and oil has

moved 20%. So... Well, are you trading these new perpetual futures? Has that become an interesting market for you? Is there enough liquidity? A lot of talk, certainly from the backers of these platforms, whether they're...

owners of hyper liquid tokens or they're the venture capitalists behind Kalshi. They're very bullish on perpetual futures and saying that that's what everybody's going to trade. But I'm interested in your opinion as practitioners, whether you think these markets are deep enough, liquid enough and whether they trend enough. Yeah, we looked at that. For the betting markets, they are not liquid enough. That's my opinion. At least now I know that Cauchy has done some block trades, larger ones, but overall, there's still more deep liquidity in that. And it's not for the fund, for bigger private portfolios, maybe you can trade them. The perpetual song, for example, I can have a liquid. We have just talked about that yesterday, actually, more recently myself, because it's very interesting that I trade them for fun, for testing, essentially. They are liquid to some extent. They have launched a lot of commodity markets, which I find very interesting because it is something that it's moving away from only having bad children from crypto. You know, that is kind of like currently now, you know, maybe having a bad stretch next day. Except for hype. Except for hype. Yeah, hype has done everything right. They have launched a token with the massive community support, essentially, and have people on the platform, which was great from the beginning. But they have also drifted away a little bit from only having crypto. I think for them, it's also important to diversify to other asset. And I think also this gives the platform a completely different meaning because now we see like soybeans and wheat and so on. They're straight. And those two, for example, have basically no liquidity. Natural gas is there has metal metal liquidity. There's huge, huge funding spreads actually at the moment in some of these markets. But there are also some which are very liquid like the equities. They are a single stock of future petrials that you can actually trade on most of your tech stock like the BDR, Mitron, stuff like that. And they are liquid. I wouldn't say you can put millions in them easily. They will be slippage and stuff like that. But still, it's fairly simple to trade them. It's fairly liquid to trade in for a larger portfolio. And it's the direction where I think awesome markets are headed because actually with these perpetual futures, you're allowing a new asset class to take rain also in the commodity space

because you're getting rid of the rolls. You have kind of like constant maturity maybe which hasn't existed before. You have a different product traded for maybe a different audience but also an older audience. For example, all the stuff I find very interesting as well, just opens up a new space to trade that stuff, a new audience as well and also a lot of efficiency because you're getting rid essentially of a lot of the friction that is maybe

associated with a classical fund, right? If you look at hyperliquid, you can launch vaults on their platform pretty easily. You can get those funded. You can trade the products and you do not have to deal with kind of like brokerages or administrators or anything between your clients come from everywhere. Let's put us away the legal side of things because this is of course like highly unregulated. We as an investment manager would not launch a product for clients that way because we wouldn't be able. just from a legal perspective to do it. But I definitely see that moving in the right direction. I think if the conventional exchanges like C&E do not launch perpetuants or competing products at some point, then the new competitors like hyper liquid will move into that space and they will siphon out some of the liquidity, also some of the AOM that might be deployed in these products. How do you determine whether there is enough liquidity? It's still hard for some. It seems to be the 24-hour volume that is mentioned there, which is kind of like a good measure. But if you trade in bigger size, you definitely have to look at the order, figure out what the spread is. For example, just look at it. Only I think the soybeans or whatever, weak markets, they had a relatively high volume. But the spread was massive, for example, in the order book. And there was also no movement in there. And then you could see that actually the funding was around 300% per year. So this already tells you that probably there's kind of like a disbalance there. And there's not enough interest on enough people taking position on the other end. I think this will get better with time as people enter the space and maybe market makers also trade on these platforms because they are already offering kind of like the the incentive to do a carry trade, for example. I mean, it's not in our range of product what we would offer, but we have done that before, right? Maybe five, six years ago, the yield on Bitcoin carry was also 40%, right? When we did trade, that was around the time when the contracts on the CME got liquidity, right? And the unit thing that you did was like holding some Bitcoin and then shorting the CME

futures against that. And That went down from 40% annualized year two, maybe three, four percent, right, very conservative because it got some adoption. And I see that I expect the same thing to happen, actually, with the other commodity contracts as well. I should add maybe real quick that Moritz is talking from a prop trading PA perspective. Yeah, we're not. Yeah, we're trading CME Bitcoin futures and CME Ethereum futures in our fund at Tuckay Capital, but we're not.

Let's talk a capital active on any of the blockchain based exchanges. Do you think that you are going to start to see these types of futures products come to CME, ICE, these other exchanges? I was talking to somebody who says that the role trade is one of the most profitable trades for these exchanges. And if they don't have any role trade, they're not going to make any money. And so their business is essentially dependent on the rolling of contracts. Is it likely that we're going to see that coming to these major US exchanges? I don't know. It's impossible for me to have a forecast on that. But in and by itself, the perpetual future is an interesting instrument, right? Yes, I mean, you would no longer have a roll trade. It's kind of like your broker giving you a margin loan. to buy a security on margin, right? You're buying the S&P 500 or the spy ETF on margin. That's also kind of like a leverage, gives you that leverage position, which never expires, but you pay funding spreads. And the same is true for that perpetual future, I presume. What people need to get comfortable with is the fact that it is decentralized. It is whatever it is, Tether or USDC, and you're on a blockchain, right? There is no CME legal person that you can send a letter to and complain if, you know, you lose your coins or tokens or you get hacked and stuff like that. I mean, that's... I don't think the institutional level is going to be on hyper liquid anytime soon. But back to what you actually do. Do you call it high octane? Obviously, there's the 8 to 12 volt that most people sort of end up around on the institutional side. You guys are more on the 25 to 30. Is that why it's high octane or are there other aspects of it? The higher volatility is the synonym for, say, high octane, right? There aren't that many high octane or higher volatility trend following funds around these days. Some, we're not the only one, but it's a minority. And the majority of funds have low volatilities. And to answer your question with, you know, what is the best? And obviously you would like to hear it's ours. That would be very arrogant, very naive and also incorrect to answer that way because

really there is. I mean, at any given point in time, there is the best, right? But it's dynamic and changing. It depends on so many factors. What we focus on is.

robustness and resilience, so we do like putting together systems, models, trading strategies that can stand the tests of time and the many curveballs that the markets throw at us, which happens repeatedly. One way of doing that is by not actually overthinking it, by not getting trapped in complexity and putting layer on top of another layer, on top of another filter and another parameter. you know, creating an over-optimized back test, which is very easy to do in today's world with the computing power that we have. Now, designing these systems in... I don't want to say simple isn't the right word, but in a robust way that strips away many of the complexities. That is actually an edge. And finding these systems isn't necessarily easy. So, yeah. That is what we believe is good for us. What we believe works for us and our clients and other trend following managers will have a different opinion on that. They will believe that their system is better suited to do this. But by and large, a well-designed long-term trend following system, they're all getting you into the same trades in the same markets at some point in time. It really depends on, yeah, do you let the trades run? How do you size them? Or are you micro tinkering with these positions while you have the position on? And that's not what we do. You're letting the trades run. So if a trade were to be put on, your model tells you what the size is, and it doubles from there, it's going to run at double the size. Is there any point? where obviously this is a good scenario, right? Where you're having a trade continue to get larger and larger. What's the biggest that an individual trade has ever gotten above its initial size? Off the top of my head, I don't know the number. I can give you some of the recent examples. I mean, cocoa is maybe a good one. But look, a market that is trending. and say we have a long position, which was the case in Coco a couple of years ago, right? And that then is a trade that deserves its place in the sun. It deserves to be large. It deserves to be moving the needle. It deserves a larger footprint in our portfolio because that's the outlier trade that's working. That is the trade that is very likely going to be very uncorrelated to other things that you have in your portfolio. It is a trade that sits on open trade profits and the volatility

is produced by a trade that is profitable and not by a trade that is producing losses. So it has all the arguments for why it deserves a bigger seat at the table than a market that hasn't moved or isn't moving. And why should we be forcing our portfolio or the systems through dynamic position sizing to rebalance the portfolio such that the trade that deserves its place in the sun, which is coconut, my example, becomes smaller, which is a counter trend or mean reversion trade that you're hiddenly executing. And other trades that haven't or other markets that aren't trending, that aren't doing well, that therefore as a result aren't as large in our portfolio, why should we increase them? It's kind of like adding to losers and taking away from winners, which is the exact opposite of a trend following strategies that is keeping losses small and letting winners run. But yes, I mean, it does come with that. So yeah, inconvenience at the point when, you know, you have this big outlier, you captured it, it has made you a lot of profits. It's now a very large position. It's a big component of your portfolio and eventually it will reverse. And it goes the other way. And that is, you know, at that point, that is also at the beginning I mentioned, you know, with the 100 basis point example. This could now be a position where we're losing much more than a hundred basis points. And we will be losing much more than a hundred basis points in that extreme example on that give back on that reversal. Like, you know, Coco goes from its high of 13,000 per ton to where it is now, which is around 4,000 or even below, you know, 4,000. So, yeah, on the way from 13,000 to 9,000, which is the 4,000 move, it will. You will experience that pain because you're giving back open trade profits and that's going to be more than 100 basis points But it's on a winning trade. It's not going to become a loser. It's going to stay in the absence of gaps and discontinuities It's going to stay a winning trade. So you don't really have to be worried too much about it Well, that is one of the interesting things about trend following that I think differentiates it from just about every other strategy out there is that it almost definitionally is not trying to be perfect. You can't catch the bottom because you can't have a trend on a reversal and you can't catch the top because a trend has to can only end after drawing down. So every other strategy, whether your value or growth, I mean, you're trying to catch these things as early as you possibly can and you're trying to sell them as late as you possibly can mostly and trend following.

just says that that is not what you're trying to do. And it's always been one of the key differentiating factors. Why not try and sell closer to add some sort of discretion? Have you looked into whether you would have any timing skill? I'm sure in many cases price leads narrative, but eventually you start to get the narrative comes through and you start to understand the fundamental drivers. of why a trend exists. Why not layer in some of that fundamental information? Let me use the exact same example again. I think it might be helpful to work through this. Coco starts at 2,500. It goes from 2,500 to 3,500 on a chart as a discretionary trader. When you look at that, it will look like a big move. And it is a big move. And you will be tempted to call that the high because the past 10 years of the past 20 years, Coco really hasn't done that much. It's kind of like jumped around and all of a sudden it had made that move and you're sitting on these open trade profits and there is this urge to now close the position, click on the mouse, realize the profit and feel good about it. But you don't know because you can't forecast that. it goes to 13,000. So with the benefit of hindsight, you look quote unquote stupid to have sold it at 3,000 or 3,500 because he could have sold it at 13,000. So because it's impossible to forecast where these markets go and where these prices go, there is no better way in our opinion to just let the system do its thing and have the system elect the exit and the exit will be on a reversal. You just have to go through that give back period where you're losing some of your open trade profits, but the system is becoming clearer and says, okay, now it's time to exit the long position and maybe turn short or just close the position and go flat, but the bullish trend is now over. As a discretionary trader, you will... Your emotions psychology will very often trick you into touching a position that shouldn't be touched Well, I guess let me use a counter example. Let's take cost me futures, right? It's clearly being driven by SK high next we have this information you could then layer in information about Okay, what memory supply is coming online and and say how much further could this thing go?

When are we actually going to get new supply into memory? Because that is what is almost likely going to signal the reversal in this. Why not layer in that type of information into the strategy if, say, your long, you know, cost fee futures? Never worked with integrating fundamental information or like our price to book or price to earnings or any news items or news flow into. into our trading systems, we believe that the information that's relevant is

incorporated in the price already, that the markets aren't 100% efficient, they're efficient enough, that prices aren't completely rational, but they are rational enough for us to make money. But yeah, the information we need there for is price only. Yeah, we definitely rely on other people doing the price fighting job for us. For example, I mean that example would be other. Fundamental managers may be bringing the price back in line, taking advantage of this location or information advantage they have or it's something because they exactly do that calculation. It would be very hard to do. I think it's, there are some macro funds, right? They are trading similar strategies that are akin to trend based on maybe macroeconomic information or sector information or company information where they use as an entry or as an exit. just very difficult to do that thing on a very large asset base. It's more usually the managers, they have maybe 10, 15 positions. They have deep research into each position. They have teams doing that all the time, right? You need to be very well informed to do that. You have to basically constantly monitor the market. And even then you can be wrong, right? It can be incomplete information that you have. It doesn't mean that you are always kind of an agreement with what the market thinks or where the market goes. Sometimes it just turns out differently. Sometimes there are surprises as well. It can just deviate from the rational story. And in that case, maybe the naive or the stupid approach is kind of just stick with the trend, stick with your simple rules and you end up in a very similar situation as well. Maybe you do not time the exit as well as other people and maybe your rationale behind the trend is different. But you can do that across a large number of different markets. because you don't need kind of like two persons who are constantly monitoring that. Because if you would do that, think of our hundred markets, which are ranging from, I don't know, commodities, agricultural commodities to equities, rates, FX, everything, it's very hard to do that. I mean, if you ask me, every commodity usually has very different information. It requires a separate team, you know, kind of like we...

It's all being scored game and all of that oil, of course. And these are all topics where you would need to kind of like be on top of all the information that is happening around the world. Does it exist maybe somewhere? Do funds do that? Yeah, probably they do. But the the hurdle to do that, the investment to that, to that is very, very high, I guess. And then again,

probably you are not right in 100% of the times, but you're right maybe 50% maybe a little bit more than 50 if you're good, like 55. And that's your edge. So yeah, I would say what we do is the robust approach to that without all the over case sheet. I think that makes sense, right? It is a question of scale. And if it if you don't have the scale to truly layer that in a robust way, then then you are playing a game of chance. I would say the firm that is is most famous for having layered this in most recently is a QR and Cliff Asnes has talked about putting fundamental. Signals into their trend following specifically for when price does get disconnected and yeah, I mean 189 billion in assets under management does does allow you the the team to be able to have people dedicated to doing that in a in a systematic and robust way So I will I will definitely grant you that I want to talk a little bit more about what's trending right now You know, it has been a tricky year. We have a president here in the United States, who seems very interested in causing markets to to reverse whether they're going up and stepping in and starting a war or they're going down and tweeting that the war is over. We're getting a lot of input that is driving markets and creating a choppy environment. So I'm interested in in what's trending, what's working right now for trend followers. Yeah, trend. In and of itself, a single market trend following, I should say, has had a very good start to the year. Maybe the month of March wasn't that great because you had the start of the war and there was been some change in positions. But by and large, I mean, you had big trends in the precious metal markets, especially in January. We're still long. Most of the precious metal markets definitely were long. Gold and silver were also long. Platinum were no longer long. palladium, that's a position that we have recently closed on the long side, but we do see, you know, we're short Bitcoin and Ethereum, the crypto markets aren't trading that well, or at least, you know, these two, I'm really not, I don't know about all the other ones to be quite honest. We're short cocoa. We are long the equity markets, no surprise there. There's only a few exceptions where we don't have long positions, but

pretty much all around the world, we have long exposure in equity markets. We're long the petroleum markets. No surprise there either, I think. We're short natural gas in the US. It's a market that doesn't really just continues going down really in price. So yeah, different. We're short coffee. We have a mixed book in bonds. We have a mixed book in currencies. There's

always something happening. But yeah, buy-in-large, I mean, this year has been good for single market trend following. But we're short cocoa since a long time. Again, it's a great trade. We're long copper. Long bean oil is a great one. So bean oil has been going higher and higher and higher for quite some time now. And that same market, that has become larger in our portfolio, for sure. That's an interesting one. We're still long cotton. Cotton has had a massive run to the upside in recent month. And it has now come back. It has given back some of these profits. But we're still long cotton. It's also a position that has a footprint in our portfolio. So how far can a position come back before the trend is over? It's impossible to say. It depends on the volatility of the market. It depends on the model that we're looking at, but it can definitely be more than the exemplary 100 basis points that I used to frame the discussion when we started, but yeah. Easily, you can have a 3% loss on a portfolio basis on a giveback market and more than that, even two. Getting long or short a trending market definitely falls within the classical trend following. What about the spread trades and some of the other things that you have layered into your strategies? So yeah, that is an interesting one. We do apply trend following systems to spread markets. And that has not worked well for us this year. It has worked very well for us last year. It has worked very well for us in recent years in general. But yes, so we are That's why I said like single market trend has a very good year and that is right because you've seen these major trends and like equities and gold, silver and, you know, markets that we've mentioned and already touched on now the petroleum markets. But spreads have been very erratic, like, you know, very volatile, very whipsaw-y. And that's why we've lost money trading these spreads or our spread trading systems. Now, this does not mean.

in expectation that we'll be losing money with these systems tomorrow, next week, next month, or next year. It could be quite the opposite. All of these systems, at least the ones that we trade, they go through periods of outperformance and underperformance, and that's quite natural. It's one of the most important things is to stick to systems for a long period of time, to evolve them carefully, thoughtfully. but not in an overreactionary way. Don't change your system only because it has produced you a loss that you don't feel good about in the right here right now. You have to be capable of working with these systems through periods of drawdown, even though

that's inconvenient and you haven't urged to change the system because you can always find a back test that would have done better in that recent period that has caused the loss. That is a recipe for disaster. You're throwing the baby out with the bathwater at the wrong point in time. The sample size on many of the systems we trade just isn't large enough to allow us to jump to these conclusions only because they go into a drawdown today. If they're well-designed and we have reason to believe that they have edge, they should be able to crystallize their edge in future price action. We may have to throw them many, many, many more balls for them to start hitting again. And, you know, periods of drawdown and underperformance for month and month and month are something that just happens. Well, that is an interesting topic. When does a signal stop working? How many incidents of the strategy failing to work? is is the signal dead versus in a drawdown. And do you think about optimizing a signal to give you more data points so you get a faster realization of something being no longer working? I mean, is it is it optimal to have a system that can go through a multi year drawdown before you finally get. enough signals to say, oh, well, it wasn't just in a drawdown. It doesn't work anymore. Would it be better to optimize for the number of data points so that you're working through these signals faster? Where does that optimization work? I think it depends on the strategy you're looking at. Kind of like some strategies you can definitely say, okay, stopped working because, for example, there was something driving the strategy and this has gone away.

for some reason like regulatory changes, changes to the market, whatever, how it behaves. For example, if you're taking advantage of an anomaly or something like that, then that can go away pretty quickly and you will be able to detect it as well. And then you can cut the market. For trend in general, it requires a long period. I mean, like looking back at the hormones of trend followers, even those that have been performing very well over the long term, you can see or even identify a stretch of several years, I think, like in 2014 to 2017, or even 2018, where it was essentially flat. You look at them and they... There's nothing happening. Of course, they go sometimes up and down, but more or less the whole sector has been very much flat, and there have been examples like Wheaton, for example, who cut all of their trend systems and went out of trend completely and then got back into trend again when they found out, well, maybe it was premature. to kind of

like call it an exit. So it's really tricky, especially tricky with systems where there's not a large sample size. One of the reasons why usually what we do, we treat markets the same way in the same system. So we do not fine-tune them, we do not kind of play around with the parameters too much. So we do not say, well, this, like we are trading at 200 day moving average on the S&P, but on the German DAX index, we trade 180 day. moving average in that system. Because this works better maybe in the past, that's not what we do. We basically treat all the markets for one system the same and say we treat it equally and just treat every time series as a different realization of basically some kind of financial market that is giving us these data points. And by that we can already say we're increasing the sample size and we would be able to be a little bit faster to detect actually when something is not working anymore. But if you only look at one market, it can take a pretty long time until you figure out if it doesn't work anymore. One of the examples we mentioned before with Coco, hasn't done anything for a long period of time, at least in our longer term months. Like the other markets as well, Ruff Rise is one of the examples that has been performing recently, but hasn't done much before. So it was always my kind of meto example. So if you wanted to throw something out, it would be Ruff Rise because... didn't do anything, and then suddenly it does something. So the way we trade, and that's also important, is that we can force the machine. Because as soon as you're forcing yourself to be in a market, so you're either long or either short, you have no chance to actually kind of like have no position in the market and actually kind of like discount something that does not work anymore. So we in a way, our systems naturally take care.

of things that do not work any longer because you will just not get any system if there's no trend and it does not pull your system down because no position also position for us. But in general, it is takes a long time to figure it out. You need a lot of data and it will be a decision that that kind of looking should take it might maybe wrong. Right. You can also say, well, something hasn't worked and some made works again for the spread market. It's a good example. Definitely one year for example, it's not enough because that's one of the things we are adding to the trend system because they spread trends have low to negative correlation actually to our single market trend and that makes them so attractive. And then there are years where it works very well like last year and you are outperforming

your competitors who are having a bad year and single trend. But then this year for example, it doesn't work well and you're like, well, should you still add it? Maybe it doesn't work and then. So it pops up again and it works well. So it's always a case to case decision and you definitely need to go through a few cycles to actually wait to see how it plays out. And I think your example of the 200 day versus the 180 day is a good one to go a little deeper on. So you have a signal and obviously these are you know, not real examples, but let's say the 200 day works really well for US markets and the 180 day works well for the DAX, but you don't want to, you don't want to be doing that. How do you, how do you decide which one dominates the signal? Right? Is it whichever one is a larger market, whichever one trends more often, whichever one has more higher volatility? How do you decide when you are getting mixed optimization signals and you have to just pick one? to keep this system from getting overly complex. We could keep it simple and stupid in that case. So in most cases, like the 180, 200 day, for example, it will be very similar, right? So in a range you're looking at, there will always be optimum point as long as you optimize, right? It might be 189 days or whatever. But in most cases, you just stick with... a general number that is kind of like widely accepted. For example, 200-day moving average is a very classical indicator, right? Probably no one who goes on screen and draws indicators will feel the urge to kind of like draw the 200-day and then add the 198-day indicator on top of that. So everyone has kind of achieved 200-day and maybe there's also kind of like some

Self-fulfilling prophecy around that the people if they are using 200 day indicators, they actually driving trends on that time frame might be true as well. But in general, we tend to stick to simple numbers, round numbers, for example, 200 day, 150 or something like that. And then it's very hard to say we do not, in the optimization, we do not focus on one point. So if you see something, for example, and maybe let's say the optimized for sharp ratio, but if you do that, and you plot your results of the various parameters versus your sharp ratio, you will identify single spots where it works really well, right? The point which has the highest sharp ratio across all the assets, for example, and that could be a parameter optimization. What we instead are looking for is a kind of like maybe a lower point. It doesn't have the highest sharp ratio, but it's very robust in terms of like, if you go down 10 or plus 10 in terms of like the next 20, 10 or 90 days. it doesn't really matter it's

just like very robust you can go to 160 maybe nothing much changes like maybe you're getting one more entries here or one less entry there but in general not a big difference of course there are kind of like boundaries to that as well you know like if you go down to 30 days of course your signal gets much faster you're getting much more entries and exits but in general look for the most stable plateau of the parameters you can find And in there, there will be your solution. And if there's kind of like a widely accepted number, and that's 200 days, then probably you stick with that. OK. So you said you're not optimizing for SHARP. What are you optimizing for? So we have no optimization running. We do not look at one number at all. I mean, we have SHARP there as a reporting number. But overall, what we want to see actually is also a very classical statistic in terms of change. Following is just the number of R multiples. So kind of like. How well do our outlier trades perform? Do we cut losses? We have a certain expectancy in terms of what we're expecting to lose on a trade. Do we see that? Or do we break through that? So do we exit on the right stock? And then on the other hand, how far can trades run? Are they able to produce these outlier-sized returns? This is more something that we conceptually look at. from a system design if we are talking about trend following trades. And of course, you will look at drawdowns, for example, or you will look at margin to equity in the second step where you kind of like design the system and you want this to be in a certain range. For example, you could have a very great outlier return strategy, something producing great trades, but it has an 80% drawdown. Yeah, surely there is not kind of in your wheelhouse. You want maybe something that is more aligned.

with the overall level of volatility that you're targeting or the overall portfolio risk. And in the end, all of these parameters need to align to a certain degree. They need to make sense. And very often when designing decisions, you can see it pretty early that one of these things is not behaving in line with the other. So you can easily see when you tend to over-optimize that one of the parameters is kind of like an outlier. and does not behave very well with the others, and then it already gives you kind of the feedback, or this does not seem to be robust, and you shouldn't expect this to play well together. The easiest example is position size. For example, when it's mentioned, you can risk 1% of your close straight equity on a position, and you can of course kind of like play around with that and scale

the buff and down. and you will immediately see that your trade distribution varies wildly and you will get very interesting results and at some point just your system will blow up. Of course, you're risking too much. If you have a certain number of trades and expectation that you have and you're risking 2% on each trade, your blow up risk is basically 100%. This is the most extreme case, but it tells you also that in between there must be something or... a position or a size, a number, which kind of like gives you your desired result also in terms of what you achieve on outlet tricks, but also draw. So when you're designing the system, if you're not optimizing for sharp, you're not really optimizing for anything, is it just does this produce, do our losses look like they're supposed to look? Do our winners look like they're supposed to look? You're building a system that gives you results that you know over time are going to produce positive returns. Is it simply that? Yeah, more or less. We look at numbers, of course, we have kind of like, we could use parameters, we could take sharp, we could take satino ratio, something that maybe is more tuned to the way we trade, that kind of like does not penalize the outlier trade. And we could put that into an optimization writing that the algorithm completely decided. What do you want to do? How much is it good? Is it bad or something and optimize for that and kind of like find two parameters? But for us, it's more really does it look robust and does it reduce the desired distribution of returns that we want to look at? It's much more important. Like you said, do the losses like what they're supposed to look like? Oh, yes. I mean, we don't want to have large losses, right? We fully expect on the trend following system to have more losing trades than winning trades we want to have.

Looking at the distribution, we want that to be positively asymmetric. We want to have some of these winning trades to be big winning trades. That's what you want to see. There are some statistics that you can look at that will tell us, okay, this is a good enough system to work with. I think what's very important is the diversification that you're getting from the markets that you're putting in. That is an exercise in and by itself. putting together a portfolio of market that works well in harmony together. You don't know that or when you're going to have a position, but you kind of like want to put yourself up such that you can have potentially positions in a very diverse set of markets both long and short. Diversification almost feels like a dirty word in this current market environment. Everybody wants

concentrated exposure to a handful of themes. You produce a strategy that by and large has been sold for its diversification benefits. There are certainly absolute return arguments to be made for trend following as well. But by and large, when you talk to people who engage in trend following, they love to talk about, oh, this strategy, when you pair it with a traditional equity portfolio, a 60-40 portfolio, look at the benefits it provides to the portfolio as a whole. And so I'm interested in whether that is how you position your strategy to your investors, is that what they're looking for? And does the choice to go for a higher volatility, is that at all impacted by what your investors are looking for from this strategy and the role it plays in their portfolios? So the answer to the last part of your question is, yes, our investors are interested in a higher volatility fund, else they would not be invested with us. Some of these investors have actually requested such a fund. Some of them are very used to investing in high volatility or higher volatility trend following funds, and they have other managers that also have higher volatility, and they're invested with these folks, and they like it for their long-term return potential. They're okay with the volatility. They're okay with the drawdown. They know what they're getting into. So these are the types of investors. We don't have institutional investors. Our volatility would be too high to appeal to an institutional allocator. They're usually not in that space with 25 or 30% fall. Why not? We can just make it smaller. They can make it smaller, but then it may not be worth their while in their time. In the time underwriting.

to underwrite and go through the operational due diligence and you know they want to then also if they are very big institutional firm they want to invest a very large sum of money or you know 100 million 200 million 300 million something like this per ticket um there's career risk at play you know usually they're in a paid position they have to report if not to a boss and to the board you don't want to look stupid you don't want you know to have anything go wrong Obviously, if you're trading at a higher level of volatility, you increase the probability of something in quotes going wrong, which is draw down or loss, right? It's not necessarily wrong. It's something different than wrong, but it's the draw down or loss. Nevertheless, an allocated doesn't like that. They like smoothness. They like high sharp ratio. They like steady eddy, something that is not giving them a negative surprise. So yeah, so this is a different thing. Now, higher

volatility futures trading because futures have inherent leverage. Trading at a high level of volatility is actually something that we believe is capital efficient. It's not just our belief, it is actually capital efficient, you know, to do that. So it's an attractive niche for us. There is a big crowded red sea of funds out there that have a lot of assets under management that trade at a lower level of volatility and they compete against each other. For big institutional allocator tickets and there is a much smaller subset and group of higher volatility trend following funds and we belong to them that I don't want to say that market is it is not competitive it is competitive but it is a relative to the alternative it is an unattractive niche at least for us and that's why we're in it. Everybody likes to say what they do is totally unique and I find that it's it seems like an attractive sales pitch to say that we do something that nobody else does. But that means nobody's shopping for it, right? And I think it's much better to position yourself as a fund that does something that people want. It might be a smaller subset of people who want this other thing, but it's something that there is a defined market for and prove that you do it well, that you do it as good or better than your competitors rather than trying to say, Oh, they're over here on this island, and we're over here doing something that's completely different. Very good points. Even if you could prove it to be true, it almost is a negative for your strategy that nobody else is doing it. Yeah. Obviously, we're...

We're not immune to this, but I completely get what it is you're saying, and it's a good point. You have these pitch books in the presentations, and they're full of uniqueness and full of alpha and full of uncorrelated and full of extremely valuable and full of very diversified and go down the list because you want to present yourself in a positive light, obviously. Now, a longer term or long term single market trend following manager. They aren't going to be that unique. If you run a long-term trend following system on a portfolio of markets, you're going to have positive and relatively high correlation to your peers. So your uniqueness kind of like goes out of the window right there. Like if you're not at least 0.5, 0.6, 0.7 or what it is correlated to your peers kind of suggests that you're not doing long-term trend following, right? So the expectation is that you're in there. So you cannot be that unique. And yet your investors want to know what is it that you do that's different or better potentially relative to or compared to another manager that I just spoke to. And I mean, look, one of the things

where I believe we're putting in a little work and thought is the portfolio of market, the composition of that portfolio that we're putting together, the markets that we trade, we're trading some markets that are smaller, less liquid. a little bit more off the beaten path. Some markets, you have to do some extra work to get to them. For instance, using OTC broker relationships, you cannot just trade them on interactive brokers. If that were your broker, it's not our broker, but any people use it. So that is one thing. We trade spreads in our portfolio, which is something that's ruling out that other funds do the same thing. We do it in our own way. it would be extremely unlikely. Nobody else is doing it in that way, right? Like we do it. So you have an element of uniqueness there. And then so what that then produces is something that is a little bit different to the rest of the pack. And therefore, a client might find that interesting as a diversifying property compared to other funds they have in their portfolio, definitely compared to the S&P 500 or or a 60-40 mix or whatever it is that your portfolio mainly consists of and trend following tends to be a absolutely magnificent diversifier for these types of assets. And maybe last point on this is you have some CTAs that came up with, oh, you know, we're producing the same system or we're using the same models, but instead of trading the markets that you already have in your portfolio through another CTA or you have the exposure through another trading program.

We're only going to be trading so-called alternative markets, which whatever is an alternative market, but let's just call it the non-standard markets, the markets which CTAs usually do not trade or haven't traded in the past. Propane gas, butane gas, Japanese power, stuff like that. Turkish interest rate swaps. Chilean peso. But you no longer trade the S&P 500. You no longer trade the 10-year note. You no longer trade your US dollar. You no longer trade oil. Obviously, if you do that, you can use the same models and get a very, very, very different return stream because the components are now very different. And then that is your uniqueness or your unique point. And you may then be very uncorrelated even to other trend followers because, yeah, you're just catching on to different trades and different trends at different points in time. And you're missing the S&P 500 bull market. So it's different in that way. we're not doing that. We want to trade all of these markets in the most diverse way we can. I want to close with this question. You're seeing a lot of the mixing of the trend directly

with the equity exposure, right? That has become increasingly popular, putting trend and equity together. People are saying it's more palatable for the non-institutional investor to see it together because it's just hard for people when they see the equity markets doing what they're doing right now to look at trend and see its benefits. You guys are not doing that. Clearly, you're finding a market for it. So what is it that your investors are really after? It's pretty much like old school, like futures trading used to be 40, 50 years ago. I wasn't around back then, but kind of like when all of this started even longer ago. It was all high volatility. Every future trader traded with a high level of volatility, even into the 90s. I mean, hedge funds had much higher volatility on average. Well, it was all a problem when it started too, right? It's dialed down massively since then, especially in the past 25 years. Volatility has halved in many cases. So, yeah, but we're not catering to that clientele. we really have more like the Heinederwith individual family office fund of funds who can allocate a smaller amount to our funds, but get something chunky in return. Well, we will leave it right there. Moritz Heiden, Moritz Siebert, thank you so much for joining me today. People can find more information at takaheitakehe.capital. Thank you guys so much, and we'll do it again sometime soon. Thank you, Max. Thanks, Max.